

AI Systems Performance Engineering

Chapter 2 — AI System Hardware Overview

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What This Talk Is About — and Why

In plain English

Modern AI models have grown so large that **no single GPU** can hold or run them. This talk explains the **hardware NVIDIA built** to solve that — the Grace Blackwell superchip and the **GB200 NVL72**, a whole rack that behaves like *one giant GPU*.

- **The problem:** trillion-parameter models are too big, too slow, and too power-hungry for one chip.
- **The challenges:** four “walls” — *memory, compute, communication, power/thermal*.
- **The motivation:** at ultrascale, tearing down each wall saves time, money, and energy.
- **The innovations:** how each NVIDIA feature knocks down a specific wall — flagged on every slide.

Why now: this is the hardware our own training & inference workloads run on — understanding it is how we make them fast and affordable.

Agenda

- ① The Problem & The Challenges
- ② The CPU and GPU Superchip
- ③ The Blackwell GPU
- ④ Ultrascale Networking: Many GPUs as One
- ⑤ Density, Power, Cooling, and Operations
- ⑥ Roadmap, Takeaways & Designing for the Machine

The Problem: One GPU Is Not Enough

The workload

A frontier model is **trillion-parameter scale** — its weights alone are **hundreds of GB to terabytes**, far bigger than any single GPU's memory, and it must be trained on exaFLOPS of compute and served in **real time** (tens of ms per token).

- No single GPU can **hold** the model, let alone run it fast enough.
- The naive fix — “**just add more GPUs**” — creates a new problem: the GPUs must **constantly exchange data**, and a slow interconnect becomes the bottleneck (Ch. 1: *the bottleneck is moving data, not FLOPS*).
- Packing that much compute in one place **strains power and cooling**.

The goal of the GB200 NVL72

Make **many GPUs behave like one giant GPU** — big enough to hold the model, fast enough to feed it, and cool enough to run it. *This chapter is how NVIDIA gets there.*

Four Walls: The Challenges This Hardware Attacks

Wall	Why it hurts (plain English)	NVIDIA's answer in this chapter
Memory	The model won't fit in one GPU's memory.	Bigger HBM, Unified/Extended memory (EGM) , dual-die, NVL72's pooled memory.
Compute	Need far more math per token, affordably.	Tensor Cores , Transformer Engine , FP8/FP4.
Communication	Split across GPUs, they wait on each other.	NVLink-C2C , NVLink 5, NVSwitch , SHARP, GPUDirect.
Power / Thermal	~ 130 kW in one rack — air can't cool it.	Liquid cooling , robust power delivery, DCGM monitoring.

*Every innovation ahead is an answer to one of these four walls — watch for the “**Solves the ... wall**” note on each slide.*

NVIDIA's Big Idea: Turn Many Chips into One Machine

*The through-line for the whole chapter: the next slides all chase one goal — make **72** GPUs behave like a single, giant, well-fed GPU.*

- **Break the memory wall** — *give the model room to live.* Stack more fast HBM, borrow the CPU's memory (unified/EGM), fuse two dies into one GPU, then *pool* the memory of all **72** GPUs so the model simply fits.
- **Break the compute wall** — *do the math cheaper.* Tensor Cores and the Transformer Engine do AI math in tiny number formats (FP8/FP4), getting far more work per chip and per watt.
- **Break the communication wall** — *stop the waiting.* NVLink, NVSwitch, and SHARP wire the GPUs together so tightly that talking to a neighbor GPU is nearly as fast as reading local memory — no slow network in the middle.
- **Break the power/thermal wall** — *keep it cool and fed.* Liquid cooling and heavy-duty power delivery let one rack safely burn **~130 kW** without melting or throttling.

*Same story every time: pack more compute closer together, then feed, connect, and cool it so it runs as **one** machine.*

GPU Terms in Plain English (Glossary) I

Cores & execution

- **SM** — a GPU's worker unit ("core"); a Blackwell GPU has hundreds.
- **Thread** — the smallest piece of work; a GPU runs thousands at once.
- **Warp** — a group of **32** threads that run the same instruction in lockstep.
- **SIMT** — one instruction drives many threads at once — where GPU speed comes from.
- **Latency hiding** — while some threads wait on memory, others compute, so the chip never sits idle.

Math & precision

- **Tensor Core** — a unit that multiplies matrices extremely fast — the engine behind AI math.
- **Transformer Engine** — auto-picks a number precision per layer to stay fast without losing accuracy.
- **FP16/FP8/FP4** — how many bits store a number; fewer bits = faster & less memory, slightly less exact.
- **FLOP / exaFLOPS** — one math operation; exaFLOPS = 10^{18} of them per second.

Memory

- **HBM (HBM3e)** — ultra-fast memory stacked next to the GPU; holds the model's active data.

GPU Terms in Plain English (Glossary) II

- **LPDDR5X** — the CPU's memory — much larger but slower than HBM.
- **Cache / registers** — tiny, very fast memories near the cores (the “memory hierarchy”); keep hot data high in it.
- **Unified memory (EGM)** — CPU + GPU memory seen as one pool, so big models can overflow into CPU memory on-package.
- **Cache-coherent** — every part automatically sees the latest value — no manual syncing.

Packaging & links

- **Die** — a single silicon chip cut from a wafer.
- **MCM / dual-die** — two dies packaged to act as one bigger GPU.
- **NV-HBI** — the very fast link joining Blackwell's two dies.
- **PCIe** — the standard, relatively slow link between CPU and add-in devices.
- **NVLink / C2C** — NVIDIA's fast links: GPU↔GPU (NVLink) and CPU↔GPU on-package (C2C).
- **NVSwitch** — a switch chip connecting every GPU to every other in one hop.
- **Bisection BW** — total data rate if you cut the system in half — a measure of fabric strength.

GPU Terms in Plain English (Glossary) III

Multi-GPU software & networking

- **Collective** — an op where many GPUs combine data (e.g. summing gradients = all-reduce).
- **NCCL** — NVIDIA's library that runs those collectives efficiently.
- **NVSHMEM / PGAS** — lets one GPU directly read/write another GPU's memory.
- **GPUDirect RDMA** — the NIC moves data straight into GPU memory, bypassing the CPU.
- **SHARP** — reductions done inside the switch hardware, so GPUs don't shuffle data.
- **DPU / SuperNIC** — smart network cards (BlueField-3 / ConnectX-8) that offload data movement & security.
- **InfiniBand / Ethernet** — the networks (Quantum-X800 / Spectrum-X800) that connect racks together.

Operations

- **Superchip** — one module combining a Grace CPU with Blackwell GPUs.
- **MIG** — splits one GPU into up to **7** isolated smaller GPUs.
- **DCGM** — NVIDIA's tool to monitor GPU utilization, memory, temperature, NVLink.
- **CDU / cold plate** — liquid-cooling parts: plates pull heat into coolant; the CDU hands it to facility water.

The Grace Blackwell Superchip

Picture this: *two chefs who used to shout orders down a narrow hallway now share one countertop — so they cook as a single team instead of waiting on each other.*

Solves the communication wall: *The CPU and GPU used to talk through a thin straw (PCIe); NVLink-C2C is a cache-coherent pipe $\sim 10\times$ wider, so the CPU + GPUs act like one chip that always sees the same data.*

- NVIDIA packages an **ARM-based CPU** with GPUs in one module that behaves like a unified engine.
- **GH200:** 1 Grace CPU + 1 Hopper GPU.
GB200: 1 Grace CPU + **2** Blackwell GPUs.
- CPU and GPUs are linked by **NVLink-C2C** (chip-to-chip): up to **~ 900 GB/s**, an order of magnitude faster than PCIe.^[1]
- PCIe Gen5 x16 (B200) $\approx$ **64 GB/s/dir**; Gen6 x16 (B300) $\approx$ **128 GB/s/dir**.
- NVLink-C2C is **cache-coherent**: CPU and GPU always see the same values.

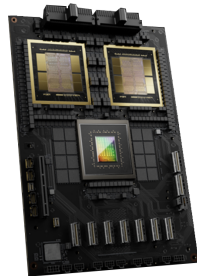


Figure 2-1. Grace Blackwell Superchip: one Grace CPU (center) and two Blackwell B200 GPUs, sharing unified memory via NVLink-C2C.

Unified, Extended GPU Memory

Picture this: a book too big for your desk (GPU memory) doesn't go to the far-off library (disk/network) — you slide the overflow onto the shelf at arm's reach (the CPU's memory).

Solves the memory wall: A model too big for GPU memory spills into the CPU's on-package memory — not slow disk or the network — so it still fits and keeps running.

One coherent memory pool per superchip

- Grace CPU: up to **~480 GB** LPDDR5X at up to **~500 GB/s**.^[2]
- Two B200 GPUs: up to **~384 GB** HBM3e (192 GB each).
- Total: **~900 GB** coherent, unified address space.

NVIDIA calls this **Unified CPU-GPU Memory / Extended GPU Memory (EGM)**.

Why it matters

- A **500 GB** model too big for one GPU's HBM fits entirely in one superchip — **no partitioning**.
- Overflow stays on-package, not on NVMe or the network.
- LPDDR5X over NVLink-C2C is **~10×** lower bandwidth and higher latency than HBM.

Keep hot data in HBM; explicitly manage placement with async prefetch and staged pipelines.

The NVIDIA Grace CPU

Picture this: *a race car (the GPU) is only as fast as its pit crew; the Grace CPU is a pit crew built to refuel and hand off data so quickly the car never idles.*

Solves the communication wall: *A GPU is useless while it waits for data; this bandwidth-first CPU exists to stream data fast enough that the GPUs never starve.*

- A custom **ARM Neoverse V2** CPU, designed for bandwidth and efficiency — modest clocks, huge memory throughput.^[2]
- Up to **~500 GB/s** to LPDDR5X and over **100 MB** of L3 cache.
- Job in the superchip: feed the GPUs so they never starve — stream data, tokenize, augment, and hand off results over NVLink-C2C.

A harmonious coupling

The CPU handles what GPUs are weak at (random access, control-heavy code); the GPUs crunch the math. The low-latency link means launching a GPU kernel from the CPU avoids the slow PCIe traversal — like a fast local call versus a slow remote one.

Blackwell: a “Dual-Die” GPU

Picture this: if you can't build one room big enough, knock down the wall between two rooms and use them as one — two chips joined so tightly the software sees a single, bigger GPU.

Solves the memory & compute wall: You can't build one die big enough, so NVIDIA fuses two to act as a single larger GPU — more transistors and $\sim 2.4\times$ the memory.

- A **multichip module (MCM)**: two GPU dies act as one unified CUDA GPU — software sees a single device.
- $\sim 104\text{B}$ transistors + 96 GB HBM3e *per die* $\Rightarrow$ $\sim 208\text{B}$ transistors, **192 GB** (**180** usable) per B200.^[3]
- vs. Hopper H100: $\sim 80\text{B}$ transistors, 80 GB — Blackwell more than doubles transistors, $\sim 2.4\times$ the memory.^[4]
- Dies talk over **NV-HBI** at **10 TB/s**.
- HBM3e bandwidth $\sim 8\text{ TB/s/GPU}$ (vs. $\sim 3.35\text{ TB/s}$ H100); **126 MB** L2 cache (vs. 50 MB).

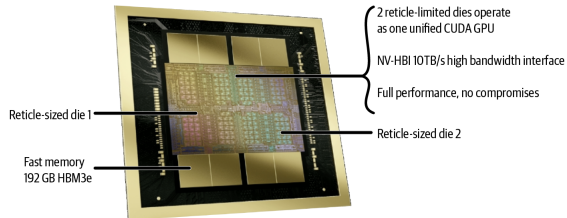


Figure 2-2. Blackwell dual-die multichip module (MCM) design.

Only **180** of **192 GB** is usable (ECC, firmware, yield); we cite **180 GB** for the B200.

Tensor Cores and the Transformer Engine

Picture this: like rounding a trip to “about 300 miles” — FP8/FP4 keeps fewer digits, faster and accurate enough.

Solves the compute wall: Do the math in smaller number formats (FP8/FP4) to run much faster and use less memory — keeping full precision only where accuracy truly needs it.

- **Tensor Cores** in each SM do matrix math at high speed; Blackwell adds **8-bit and 4-bit** floating point.
- The **Transformer Engine (TE)** mixes precision — critical layers in FP16/BF16, tolerant ones in FP8/FP4 — to preserve accuracy.
- Hopper introduced FP8 (**2×** FP16); Blackwell adds **NVFP4**, potentially doubling FP8 throughput again.^[5]
- An NVL72 rack reaches over **1.4 exaFLOPS** of FP4 Tensor Core throughput^[6]; FP4 also halves per-parameter memory vs. FP8.

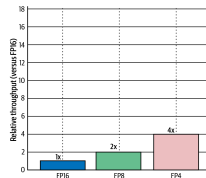


Figure 2-3. Relative speedup of FP8 and FP4 compared to FP16.

*H100 served a 1.8T-param MoE at ~3.4 tok/s/GPU; NVL72 hit ~150 tok/s/GPU — roughly **30×** faster.^[1]*

Inside the GPU: SMs, Threads, Warps, and Memory

Picture this: a supermarket with hundreds of checkout lanes: if one shopper fumbles for change, the other lanes keep ringing up sales — so the store is never idle waiting on one person.

Solves the memory wall: Thousands of threads run at once, so while some wait on slow memory others keep computing — the chip hides the delay instead of stalling.

- A Blackwell GPU has **hundreds of SMs**, the GPU's “cores.”
- An SM runs threads in **warps** of exactly **32** threads in lockstep — the **SIMT** model.
- Many warps run concurrently so that while one waits on memory, another runs — **latency hiding**.
- Hierarchy: registers → shared memory → L1 → L2 (**126 MB**, shared) → HBM.^[3]
- Keep reusable data high in the hierarchy to approach peak FLOPS without stalling.

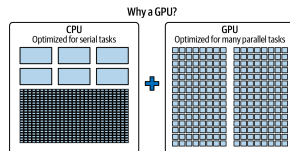


Figure 2-4. CPU cores vs. GPU cores.

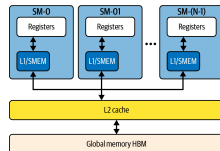


Figure 2-5. GPU memory hierarchy.

The GB200/GB300 NVL72: a Supercomputer in a Rack

Picture this: instead of 72 people emailing spreadsheets back and forth, seat them at one giant round table where everyone shares the same whiteboard — that's a rack acting as one machine.

Solves the memory & communication wall: Wire 72 GPUs so tightly they share one address space and a one-hop fabric — a single machine big enough to hold and run the whole model.

- **72 Blackwell GPUs** + **36 Grace CPUs**, all linked by NVLink.^[6]
- Built as **18** compute nodes — each node = 2 superchips (4 GPUs + 2 CPUs).
- Every GPU reaches any other GPU through the **NVLink Switch fabric** in a single NVLink domain.
- Delivered as a **preintegrated rack appliance**: cabling, power, and cooling assembled and validated by NVIDIA.
- Ships with Base Command Manager, SLURM, and Kubernetes for scheduling.

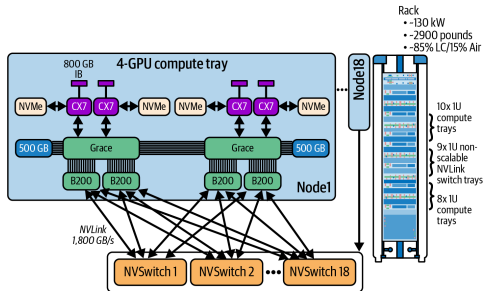


Figure 2-9. Each GPU connects to each NVSwitch — 72 fully connected GPUs (36 superchips) and 18 NVSwitches.

NVLink and NVSwitch

Picture this: *not a chain of phone-tag calls passed person to person, but a conference bridge where any GPU dials any other directly — one hop, full speed, no middlemen.*

Solves the communication wall: *Every GPU reaches every other at full speed in one hop, so splitting a model across 72 GPUs no longer means waiting on a slow network.*

Per-GPU connectivity

- Each Blackwell GPU: **18** NVLink 5 ports.
- Aggregate **1.8 TB/s** bidirectional per GPU (18 links $\times$ 100 GB/s).^[7]
- Double Hopper's per-GPU NVLink bandwidth.
- Inter-GPU latency: single-digit microseconds.

Full crossbar fabric

- **9** switch trays $\times$ 2 chips = **18** NVSwitch chips.
- Each GPU wires one link to each NVSwitch $\Rightarrow$ any GPU reaches any other in **one hop**.
- Rack bisection bandwidth: **~ 130 TB/s** — many times a top-end InfiniBand cluster.

A fully connected fabric with a global address space across GPUs, enabling efficient collectives.

Multi-GPU Programming: NVL72 vs. an H100 Cluster

Picture this: *roommates who grab what they need straight from a shared fridge, instead of asking a middleman to fetch every item — GPUs read each other's memory directly.*

Solves the communication wall: *GPUs read each other's memory directly and let the NIC move data straight into GPU memory — the CPU is out of the hot path entirely.*

- GPUs access each other's memory over NVLink via peer-to-peer / PGAS (**NVSHMEM**).
- Global address space exists, but GPU caches are **not** globally coherent — only the CPU-GPU C2C path is.
- **NCCL** and NVSHMEM supply synchronization; **GPUDirect RDMA** moves data NIC↔GPU without the CPU.

Concrete comparison

- NVL72: **1.8 TB/s/GPU**, **1–2 μ s** latency.
- InfiniBand: **20–80 GB/s/GPU**, **5–10 μ s+**.
- Collective overhead drops from tens of percent to a few percent.

Design to keep communication **intra-rack**; use InfiniBand/Ethernet **inter-rack** only when scaling beyond one NVL72.

In-Network Aggregation with NVIDIA SHARP

Picture this: *instead of every classmate mailing their tally to one student to sum, the mailroom itself adds them up on the way through — the network does the math so the GPUs don't have to.*

Solves the communication wall: *The switches add up partial results themselves, so GPUs don't waste time shuffling data back and forth during collectives.*

- **SHARP** (Scalable Hierarchical Aggregation and Reduction Protocol) runs reduction engines *inside* the NVSwitch ASICs.^[8]
- The fabric combines partial results without funneling data through the GPUs — offloading collectives to the switch hardware.
- Benefits: lower collective latency, less network traffic, GPUs freed for real compute, **near-linear** scaling as GPU count grows.

Beyond the rack

Each node has **ConnectX-8 800 Gb/s NICs**^[10] and a **BlueField-3 DPU** that offloads RDMA, NVMe-oF, and packet processing — moving data directly into GPU memory. Racks scale out over Quantum-X800 InfiniBand / Spectrum-X800 Ethernet to form **AI factories**. SHARP came from NVIDIA's Mellanox acquisition.

Compute Density and Power

Picture this: cram a whole neighborhood's worth of electricity into one closet: ~ 130 kW is roughly 40 homes' draw — packed into a single rack.

This is the Power/Thermal wall: the price of packing this much compute in one rack — the next slide is the answer.

- A fully loaded NVL72 draws up to ~ 130 kW — more than $2\times$ the prior 50–60 kW AI rack.^[9]
- Budget: $18 \text{ nodes} \times 6 \text{ kW} = \sim 110$ kW compute + ~ 20 kW switches/cooling.
- Fed by high-amperage circuits (e.g., 415 V 3-phase), often with **redundant feeds** each sized for the full load.

A mini power substation

- Transients matter: 72 GPUs ramping can spike tens of kW in milliseconds — clocks are staggered to smooth surges.
- Eight racks (576 GPUs) ≈ 1 MW — a small data center.
- Concentrating compute yields high **compute-per-watt**.

Liquid Cooling vs. Air Cooling

Picture this: you cool a fever with a wet cloth, not by fanning it — water carries away far more heat than air ever could, which is why the whole rack is plumbed with coolant.

Solves the power/thermal wall: Air physically can't remove **130 kW** from one rack; water carries the heat away, keeping GPUs cool enough to run at full clocks.

- Air cannot cool **130 kW** in one rack — each GPU can dissipate **~1,200 W**. The NVL72 is **fully liquid-cooled**.^[9]
- **Cold plates** on every superchip and NVSwitch carry heat via coolant through hoses, manifolds, and pumps, with quick-disconnect couplings.
- A **Coolant Distribution Unit (CDU)** transfers heat to the facility's chilled-water loop (**20–30°C**); warm-water designs enable chiller-free cooling.

Benefits

Keeps GPUs at **50–70°C**, avoiding thermal throttling and lowering power leakage. Roughly **150–200 L/min** at a **10–12°C** rise dissipates **~130 kW**.

Physical reality

A filled rack weighs **~3,000 lbs** (1.3–1.4 t) — floors must be reinforced. Meta, xAI, and Google now adopt liquid cooling at scale.

Monitoring, Utilization, Sharing, and Scheduling

Monitor everything

- **DCGM** tracks per-GPU utilization, memory, temperature, NVLink throughput.
- Aim for **near 100%** utilization; 50% signals a data or sync bottleneck.
- At **130 kW**, power monitoring and **power/clock caps** cut waste over long runs.

Share the rack

- SLURM / Kubernetes carve GPUs across teams (e.g., 8 / 16 / 48).
- **MIG** splits one GPU into up to **7** isolated instances.
- BlueField DPUs enable **secure multitenancy**.
- Track goodput; some use **chargeback** per GPU-hour.

ROI of Upgrading Your Hardware

- The question comes down to **performance per dollar** — if an NVL72 replaces several older racks, it can save money long-term.
- One Blackwell GPU can replace **2–3** Hopper GPUs in throughput.

Case study

100 H100s (~70 kW) could be handled by **50** Blackwell GPUs (~50 kW) — each is more than twice as fast. Half as many GPUs can be cost-neutral or better, saving tens of thousands of dollars in power per year and reducing server, RAM, and networking overhead. Payback in **1–2** years when kept busy.

Soft benefits: fewer models split across GPUs, simpler architecture, and access to the newest software optimizations.

The Arc: From Graphics Chip to AI Supercomputer

Past → present

Year	Chip	Key leap
1999	GeForce 256	NVIDIA coins “GPU” — fixed graphics
2006	Tesla + CUDA	GPU becomes programmable (GPGPU)
2016	Pascal P100	NVLink + HBM2 — DL era begins
2017	Volta V100	first Tensor Cores
2020	Ampere A100	MIG, TF32, structured sparsity
2022	Hopper H100	Transformer Engine , FP8
2024	Blackwell	dual-die, FP4 , NVLink 5, NVL72

Planned future

- **2025 — Blackwell Ultra (GB300):** +50% memory, 1.5× compute.
- **2026 — Vera Rubin:** new Vera CPU, Rubin GPU, **NVLink 6**.
- **2027 — Rubin Ultra:** quad-die, NVL576, **3–4 exaFLOPS**/rack.
- **2028 — Feynman:** 2nm, HBM5, inference-heavy reasoning.

The pattern

Graphics → programmable compute → AI accelerator → **rack-scale supercomputer**. Every generation **doubles something**: dies, memory, or interconnect.

Dates 2027–2028 are press roadmap estimates, not official NVIDIA specs.

A Glimpse into NVIDIA's Roadmap

Blackwell Ultra (GB300)

Drop-in NVL72 upgrade: **288 GB**/GPU (+50% vs. 180 GB), **1.5×** compute, **45–50%** higher inference throughput; same NVLink 5.

Rubin Ultra (2027)

Possible **quad-die** R300 module: **1 TB** HBM per module; NVL144 / NVL576 configs pushing **3–4 exaFLOPS** per rack.

Vera Rubin (2026)

Vera CPU (3nm, LPDDR6 ~1 TB/s) + 2 Rubin GPUs; HBM ~13–14 TB/s, ~**200** SMs/die, **NVLink 6**. Racks ~5× the performance — near **600 kW**.^[11]

Feynman (2028)

Likely 2nm, HBM5, more DDR, perhaps **eight dies** — optimized for inference-heavy reasoning workloads.

NVIDIA **doubles something every generation**: dies, memory, or interconnect bandwidth.

Rubin roadmap per NVIDIA [11]; Feynman/2028 and exact GB300 288 GB figures are press roadmap estimates, not official NVIDIA specs.

Key Takeaways

- ① **Integrated superchip** — Grace CPU fused with Blackwell GPUs creates a unified memory space, eliminating manual CPU↔GPU copies.
- ② **Unified memory** — coherent interconnect cuts programming complexity; developers focus on algorithms, not data movement.
- ③ **Ultrafast interconnects** — NVLink-C2C, NVLink 5, and NVSwitch give extreme intra-rack bandwidth and low latency.
- ④ **High-density NVL72** — 72 GPUs and a huge unified memory pool in one rack enable otherwise-impractical models.
- ⑤ **Cooling & power** — liquid cooling and robust power at **~130 kW**/rack keep the system reliable.
- ⑥ **2–2.5× gains** over Hopper — up to **30×** faster inference with FP4 Tensor Cores + TE, plus fewer GPUs.
- ⑦ **Software stack & a future-proof roadmap** keep doubling key parameters — codesign is the theme throughout.

Back to Mechanical Sympathy 机械感知力: Designing for This Machine

mechanical sympathy means writing algorithms that *fit the machine's grain* instead of fighting it. Now that we know the Grace–Blackwell stack, each hardware truth becomes a concrete design move.

Hardware truth (Ch. 2)	What it means for algorithm design (with examples)
Tensor Cores love low precision	Design in mixed/low precision, not FP32. <i>E.g.</i> FP8 training with the Transformer Engine; NVFP4 weight quantization for inference.
HBM is far, SRAM is near	Restructure math to <i>reuse on-chip data</i> and avoid HBM round-trips. <i>E.g.</i> FlashAttention tiles attention to stay in shared memory; kernel fusion to cut memory traffic.
NVLink is fast, the network is slow	Place the <i>heaviest communication inside the NVLink domain</i> . <i>E.g.</i> keep tensor/expert parallelism intra-rack; overlap all-reduce with compute so GPUs never wait.
Unified/EGM memory is huge	Trade a little bandwidth for capacity instead of recomputing or spilling to disk. <i>E.g.</i> offload optimizer states / KV cache to Grace memory.
Switches can do math (SHARP)	Prefer collective patterns the fabric accelerates. <i>E.g.</i> large fused all-reduce for gradients rather than many small messages.

The fastest algorithm is the one shaped like the hardware — precision, locality, and placement are design choices, not afterthoughts.

Questions & Discussion

How do we keep every flop and every byte of an NVL72 busy?

References & Further Reading I

- ① NVIDIA. *GB200 NVL72 Delivers Trillion-Parameter LLM Training and Real-Time Inference* (2024). NVIDIA Developer Blog. *Superchip = 2 Blackwell + 1 Grace; NVLink-C2C 900 GB/s coherent; 30× inference (150 vs. 3.4 tok/s/GPU on 1.8T MoE).*
- ② NVIDIA. *Grace CPU Superchip* product page. nvidia.com/.../grace-cpu-superchip. 72 Neoverse V2 cores; up to 480 GB LPDDR5X; 114 MB L3 (the 480 GB config is rated ~384 GB/s).
- ③ NVIDIA. *Blackwell Architecture overview + Blackwell Tuning Guide* (CUDA docs). 208B transistors, dual-die, NV-HBI 10 TB/s, 8 TB/s HBM, 126 MB L2, 180 GB usable.
- ④ NVIDIA. *Hopper Architecture In-Depth* blog + *H100* product page. 80B transistors, 80 GB HBM3, 50 MB L2, 3.35 TB/s.
- ⑤ NVIDIA. *Transformer Engine* (GitHub); *NVFP4* blog. *FP8 on Hopper; MXFP8 + NVFP4 on Blackwell.*
- ⑥ NVIDIA. *GB200 NVL72* product page. nvidia.com/.../gb200-nvl72. 72 GPU / 36 CPU; 1.8 TB/s per-GPU NVLink; 130 TB/s aggregate; 1.44 EF FP4.
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- ⑧ NVIDIA. *SHARP* documentation (v3.0). networking-docs.nvidia.com/sharp. *In-network offload of collective reductions. (Mellanox-origin is historical, not from this doc.)*
- ⑨ NVIDIA. *Blackwell Platform Boosts Water Efficiency by Over 300×* (2024). NVIDIA Blog. *Direct-to-chip liquid cooling: cold plates, CDUs (NVIDIA rates ~120 kW/rack; 132 kW is a partner spec).*
- ⑩ NVIDIA. *ConnectX-8 SuperNIC* (800 Gb/s) + *BlueField-3 DPU* page.
- ⑪ NVIDIA. *Vera Rubin POD* (2025). NVIDIA Developer Blog. *Vera Rubin NVL72 (2H 2026), Rubin Ultra NVL576, NVL144/Kyber; references GB300 NVL72 (2025). Feynman/2028 are press roadmap estimates.*